

國立臺灣師範大學
National Taiwan Normal University



國立臺灣大學
National Taiwan University

機器心智理論推理：訓練、測試、與應用

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09/30/2026 @ 師大資工系

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- **國立臺灣大學**資訊工程學系 助理教授（2025 至今）
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- **亞馬遜公司** 實習研究員（2021、2022 暑假）

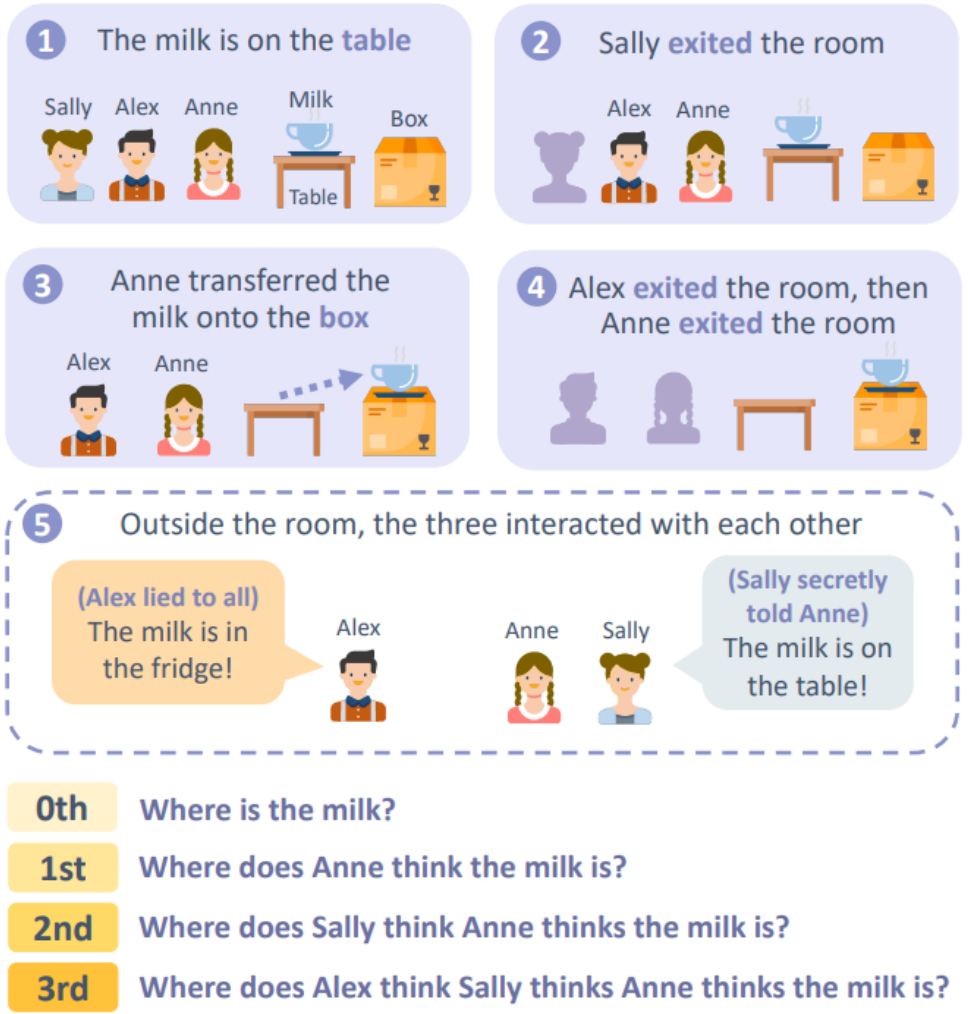
- **約翰霍普金斯大學**電機與電腦工程 博士（2019 – 2023）
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What is Theory of Mind (ToM)?

- ToM is the ability to **understand other people's mental states**, such as thoughts, emotions, intentions, and beliefs
- **Machine ToM** aims to replicate this human's innate ability in AI agents



[He et al. EMNLP-Findings'23]



ToM Promotes Effective Human–AI Collaboration

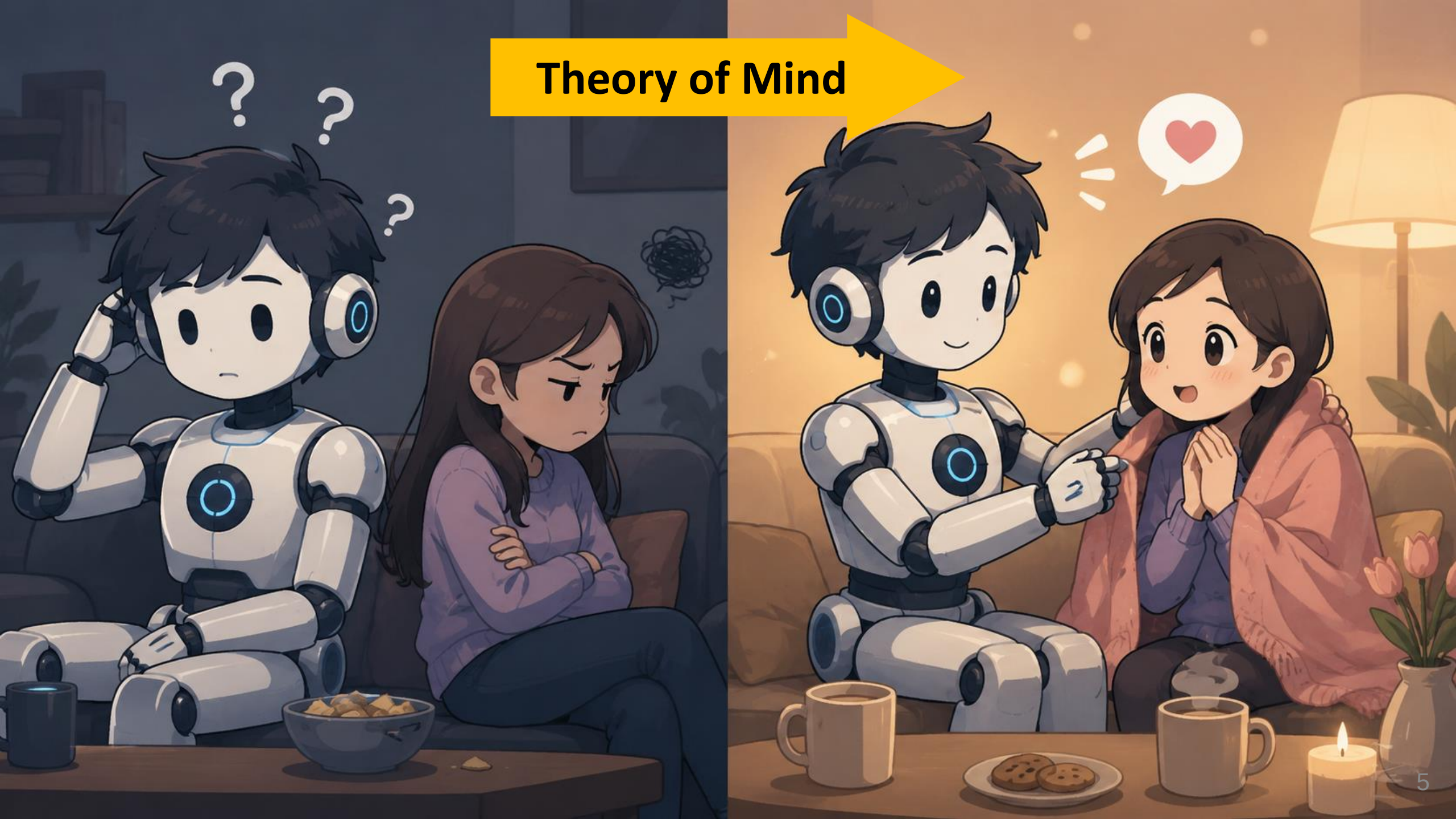
- Infer user's mental states, such as thoughts, intentions, and beliefs
- Track perspectives (what the human knows vs. doesn't know)
- Predict next actions and anticipate needs

→ More aligned assistance

→ Fewer misunderstandings

→ Safer interactions




Theory of Mind

Machine ToM Across AI Development

Training



Cost-efficient training
(ICML 2025 Spotlight)

Evaluation



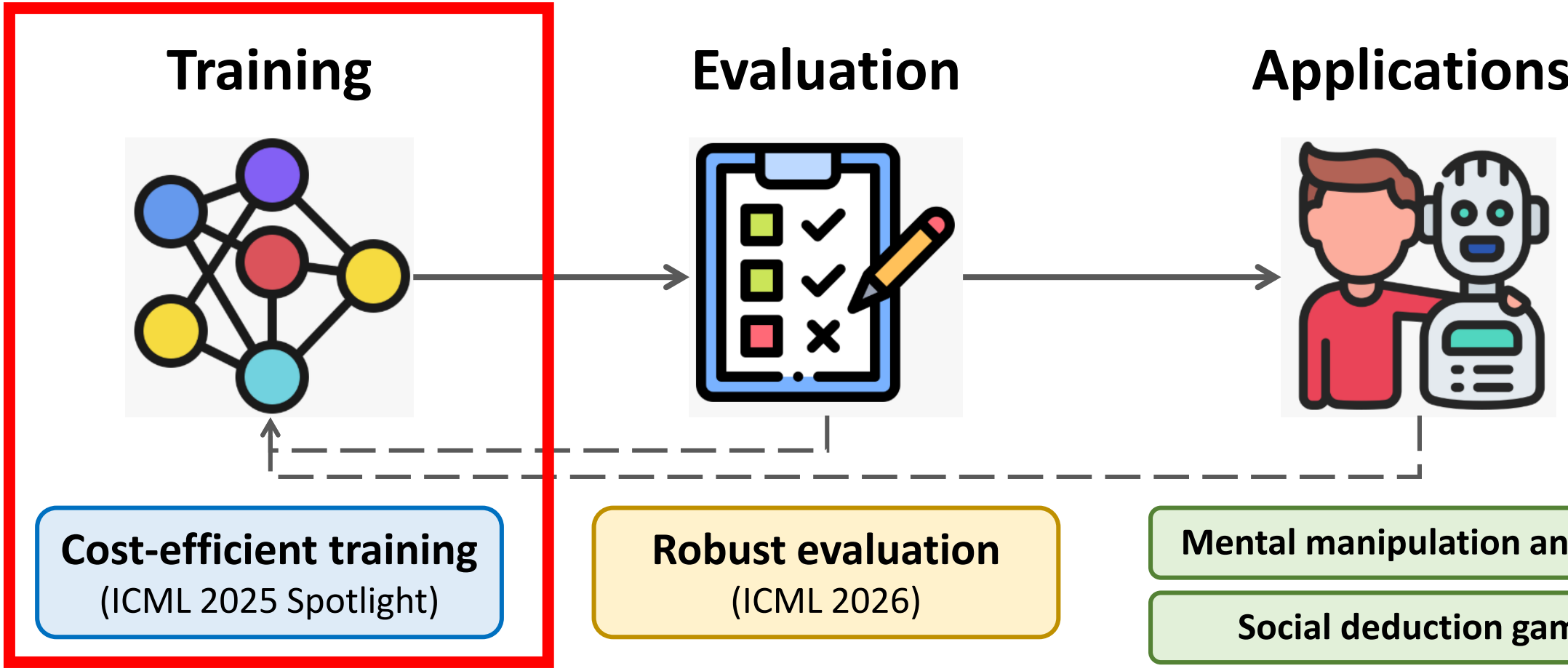
Robust evaluation
(ICML 2026)

Applications



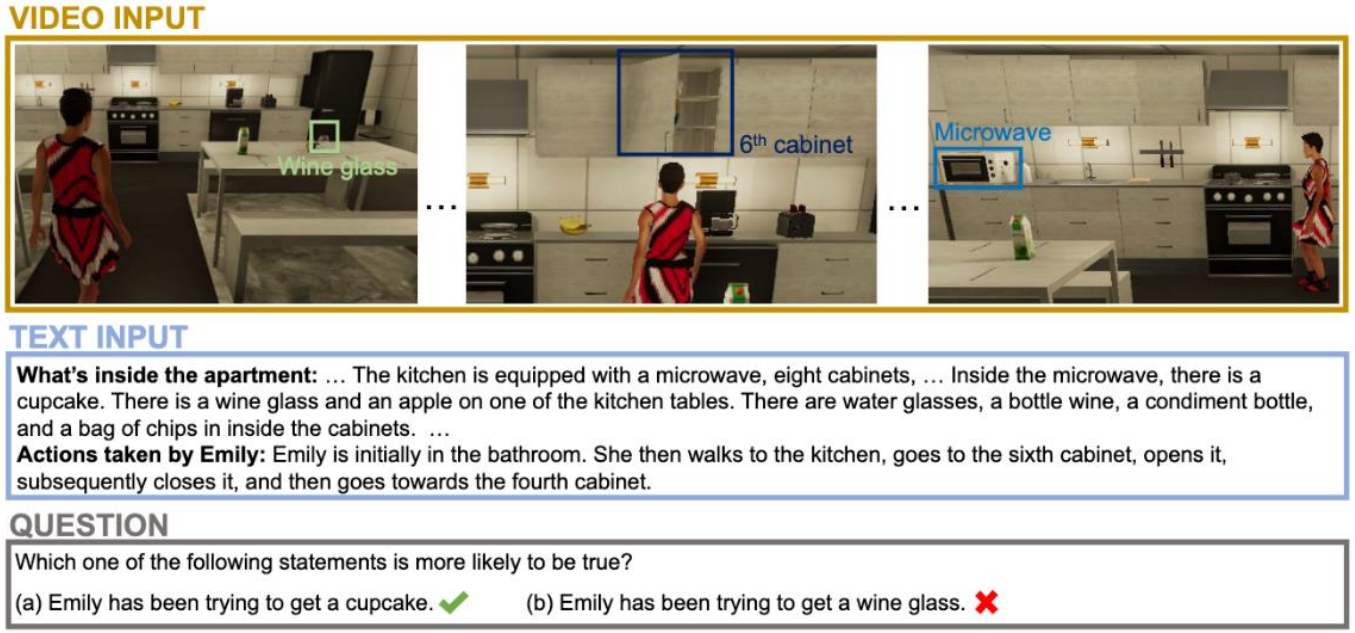
- **Mental manipulation analysis**
- **Social deduction game**
- **Autonomous driving**

Machine ToM Across AI Development



Multimodal ToM Training is Expensive

- Scaling Law: Larger models generally exhibit stronger capabilities
- However, multimodal ToM training is expensive
- How can we efficiently scale multimodal ToM training to larger models, such as Llama3.1-**405B**?



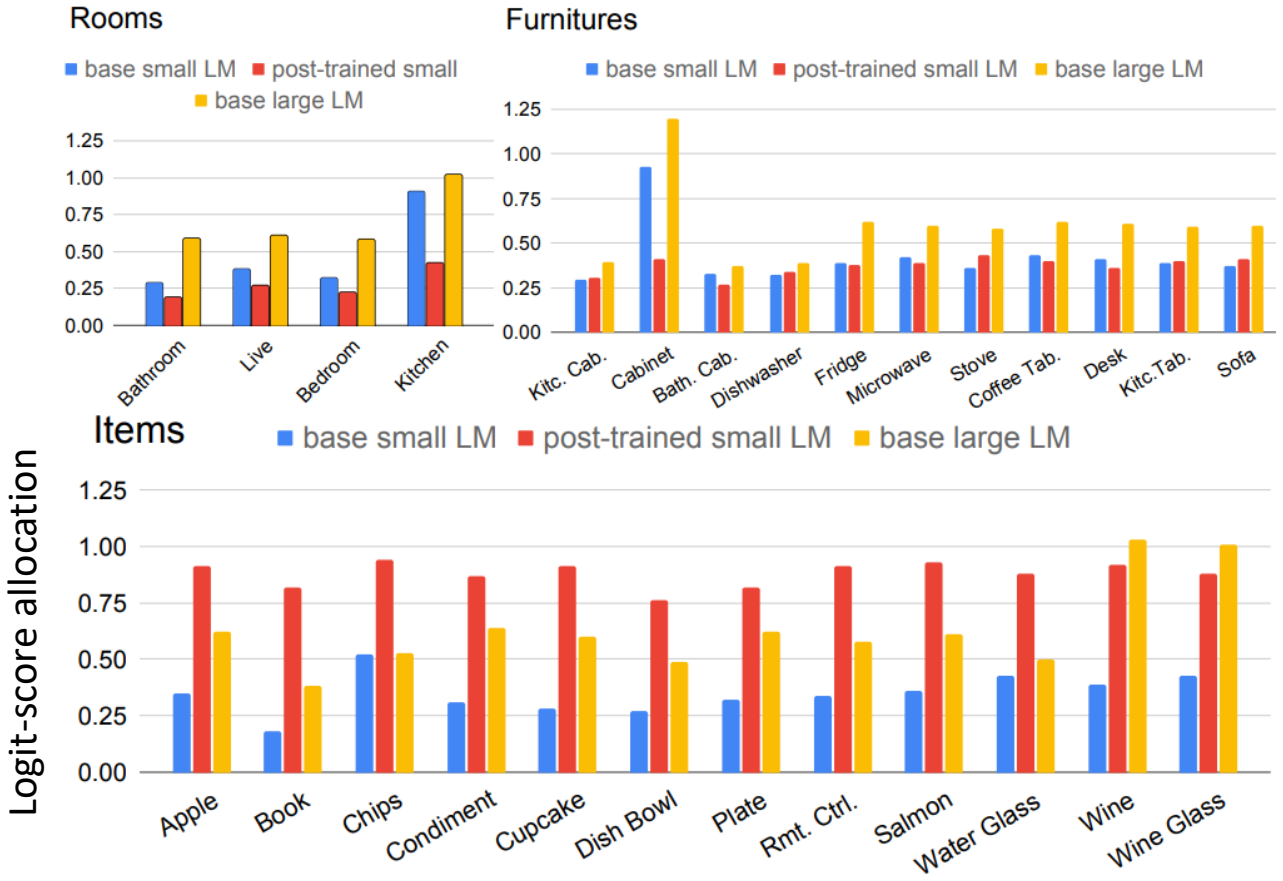
[Jin et al. ACL'24]

Model Behavior Analysis

- **Base Small LM** (8B)
  Post-Trained Small LM (8B)
  Base Large LM (405B)

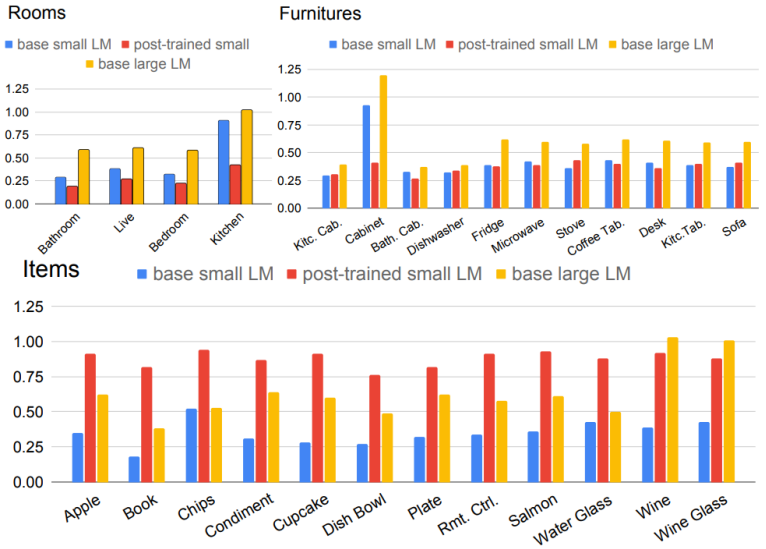
- Three levels of concept granularity:
  Room
  └ **Furniture**
  └ **Item**

How much does the model rely on each concept level during decision-making?



Model Behavior Analysis

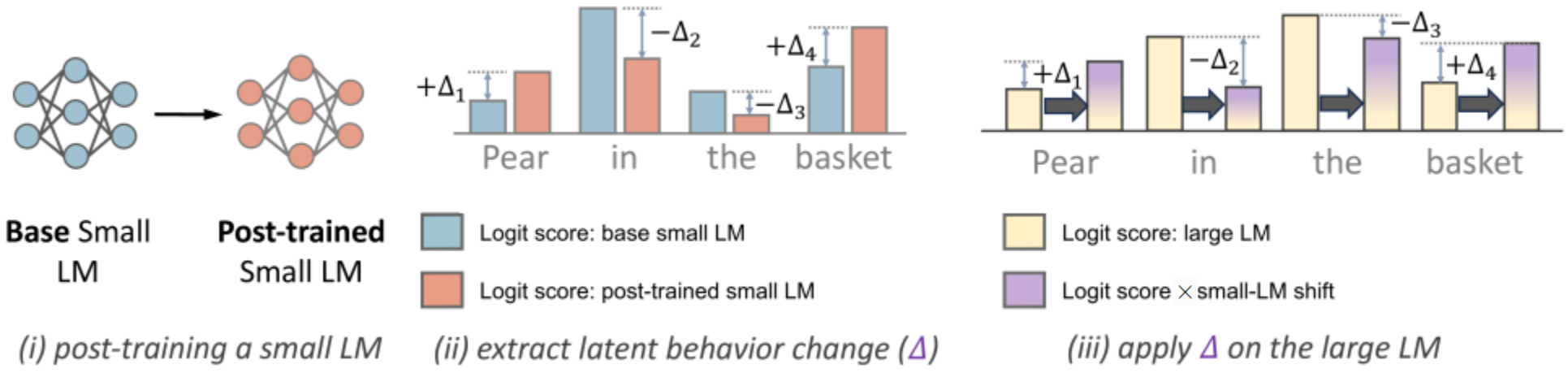
- **Post-Trained Small LM** and **Base Large LM** exhibit distinct behaviors and complementary strengths



Model	Concept level	General
Post-Trained Small LM	Concentrate on fine-grained objects (item)	Align better with specific ToM scenarios
Base Large LM	Reason over higher-level concepts (room, furniture)	Have better general world knowledge and reasoning

Weak-to-Strong Control Approach

- Transfer the post-trained alignment from a **Small LM** to a **Large LM**
- Adapt the **Large LM**'s ToM behavior by training only the **Small LM**



$$\text{Logits}_{\text{adapted large}} = \text{Logits}_{\text{base large}} \times \left(\frac{\text{Logits}_{\text{post-trained small}}}{\text{Logits}_{\text{base small}}} \right)$$

Results

- **Dataset**: MMToM-QA
- **Metric**: Accuracy
- **Ours**: post-train Llama-3.1-8B + weak-to-strong control

LM	config	belief inference				goal inference					all
		1.1	1.2	1.3	avg.	2.1	2.2	2.3	2.4	avg.	
Llama-3.1	8B-zero-shot	88.00	72.00	91.00	83.67	65.33	62.67	22.67	54.67	51.33	65.19
	8B-post-trained	90.00	71.00	93.00	84.67	69.33	72.00	62.67	72.00	69.00	75.71
	70B-zero-shot	85.00	63.00	93.00	80.33	72.00	76.00	16.00	61.33	56.33	66.62
	70B-post-trained	91.00	69.00	**95.00**	85.00	69.33	80.00	29.33	69.33	62.00	71.86
	405B-zero-shot	86.00	70.00	90.00	82.00	73.33	78.67	21.33	66.67	60.00	69.43
	70B-ours	90.00	74.00	93.00	85.67	**74.67**	77.33	70.67	76.00	74.67	79.38
	405B-ours	**92.00**	**76.00**	93.00	**87.00**	73.33	**80.00**	**76.00**	**78.67**	**77.00**	**81.29**

Machine ToM Across AI Development

Training



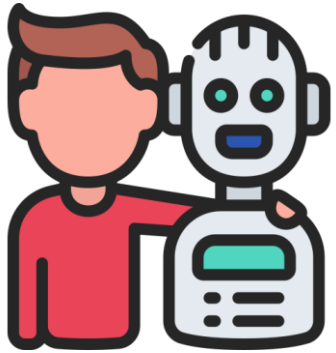
Cost-efficient training
(ICML 2025 Spotlight)

Evaluation



Robust evaluation
(ICML 2026)

Applications



Mental manipulation analysis

Social deduction game

Autonomous driving

Sustainability



Long-horizon continual adaptation

What We Observed in ToM Evaluation?

- State-of-the-art LLMs do not perform well on ToM tasks (around 70% accuracy)

Model	4th-order ToM	Hi-ToM	ToMi	ExploreToM (Raw)	ExploreToM (Infilled)
GPT-4o	46.17%	69.00%	61.96%	67.35%	62.48%
GPT-4o-mini	30.50%	58.50%	**70.64%**	**69.32%**	**66.04%**
DeepSeek-v3	**58.67%**	**70.17%**	57.17%	65.01%	64.73%

- ToM's property: long, complicated, contains causal relationships
- Intuition: use **reasoning**
- → Can we improve LLMs' ToM via **RL post-training**?

[Lu et al. arXiv'25]

What We Observed in ToM Evaluation?

- A study on **ToM post-training** appeared on arXiv in **May 2025**
- **Their key takeaway**: SFT achieves competitive performance with RL on current ToM benchmarks for ToM post-training

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Do Theory of Mind Benchmarks Need Explicit Human-like Reasoning in Language Models?

Yi-Long Lu,*† **Chunhui Zhang**,*† **Jiajun Song**, **Lifeng Fan**, **Wei Wang**†
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arXiv, May 2025

What We Observed in ToM Evaluation?

- **Their key takeaway**:
  SFT achieves competitive performance with RL on current ToM benchmarks for ToM post-training

- Interesting study

But…

(1st, 2nd, 3rd-order)

Model	4th-order ToM	Hi-ToM
GPT-4o	46.17%	69.00%
GPT-4o-mini	30.50%	58.50%
DeepSeek-v3	**58.67%**	**70.17%**
Qwen2.5-0.5B-Instruct	23.83%	30.33%
Qwen2.5-1.5B-Instruct	25.17%	40.67%
Qwen2.5-3B-Instruct	27.50%	39.17%
Qwen2.5-7B-Instruct	28.83%	52.17%
Qwen2.5-7B-Instruct-1M	17.83%	40.67%
Qwen2.5-0.5B-Instruct (RL)	85.83%	70.83%
Qwen2.5-1.5B-Instruct (RL)	89.33%	79.17%
Qwen2.5-3B-Instruct (RL)	88.17%	81.17%
Qwen2.5-7B-Instruct (RL)	82.83%	83.33%
Qwen2.5-7B-Instruct-1M (RL)	**94.50%**	**84.50%**
Qwen2.5-0.5B-Instruct (SFT)	88.17%	81.00%
Qwen2.5-1.5B-Instruct (SFT)	86.17%	80.50%
Qwen2.5-3B-Instruct (SFT)	92.67%	87.00%
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Qwen2.5-7B-Instruct-1M (SFT)	93.67%	86.50%

[Lu et al. arXiv'25]

What We Observed in ToM Evaluation?

- **Our key takeaway**:
  Why do all ToM **post-trained models** perform better on 4th-order ToM than on lower orders, while all **non–post-trained models** do not?

- Higher-orc

1st Where doe
2nd Where doe
3rd Where does Alex think Sally thinks Anne thinks the milk is?

We reproduced, same results!

Model	4th-order ToM	(1st, 2nd, 3rd-order) Hi-ToM
GPT-4o	46.17%	69.00%
GPT-4o-mini	30.50%	58.50%
DeepSeek-v3	**58.67%**	**70.17%**
Qwen2.5-0.5B-Instruct	23.83%	30.33%
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	17.83%	40.67%
	85.83%	70.83%
	89.33%	79.17%
	88.17%	81.17%
	82.83%	83.33%
	94.50%	**84.50%**
	88.17%	81.00%
	86.17%	80.50%
Qwen2.5-3B-Instruct (SFT)	92.67%	87.00%
Qwen2.5-7B-Instruct (SFT)	**94.00%**	**87.33%**
Qwen2.5-7B-Instruct-1M (SFT)	93.67%	86.50%

[He et al. EMNLP-Findings'23] [Lu et al. arXiv'25]

We Found Shortcuts in ToM Benchmarks

- In the Hi-ToM dataset, a **shortcut** can solve **99%** of the 3rd and 4th

Shortcuts lead to a false expectation about model capabilities!

of what happens afterward.

1st	Where does Anne think the milk is?
2nd	Where does Sally think Anne thinks the milk is?
3rd	Where does Alex think Sally thinks Anne thinks the milk is?

Model	4th-order ToM	(1st, 2nd, 3rd-order) Hi-ToM
GPT-4o	46.17%	69.00%
Qwen2.5-0.5B-Instruct (RL)	85.83%	70.83%
Qwen2.5-1.5B-Instruct (RL)	89.33%	79.17%
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Qwen2.5-7B-Instruct-1M (SFT)	93.67%	86.50%

[He et al. EMNLP-Findings'23] [Lu et al. arXiv'25]

We Found Shortcuts in ToM Benchmarks

- We conduct **the first systematic examination** of shortcuts for existing ToM datasets

- **(1) LLM-guided rules**: query an advanced LLM to discover potential shortcuts, which works well

- **(2) Lexical associations**: Check spurious lexical associations

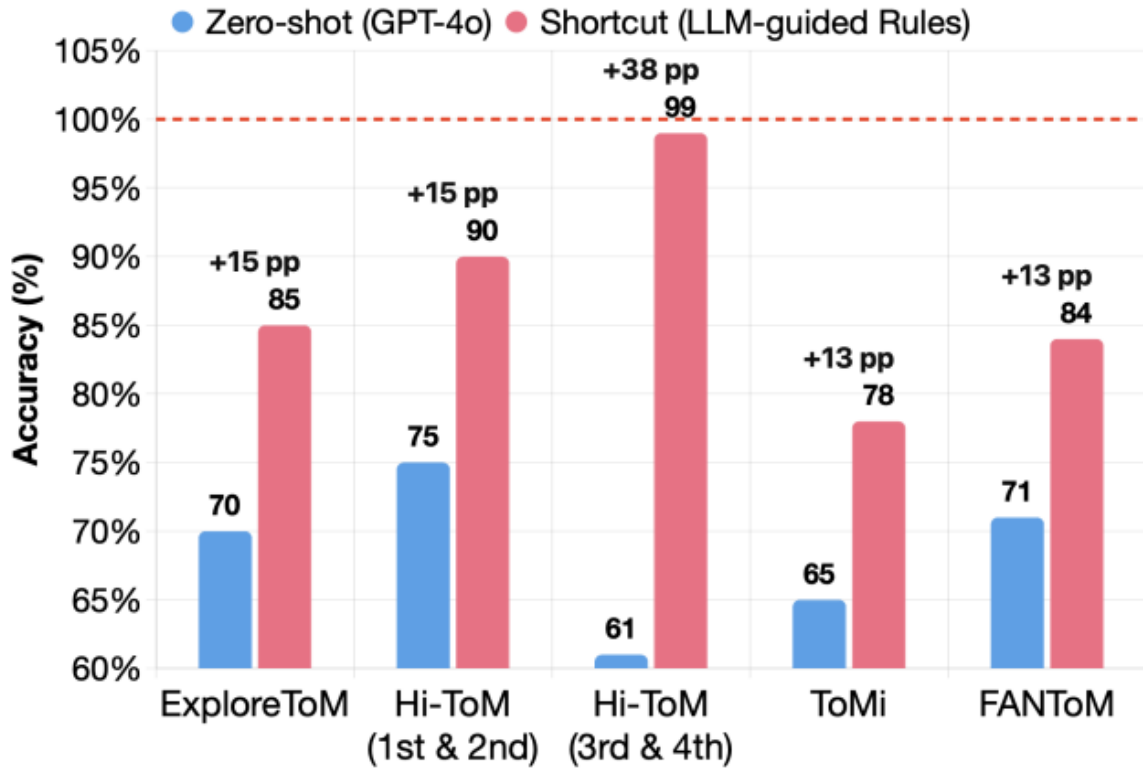
We Found Shortcuts in ToM Benchmarks

- We provide a comprehensive investigation of 8 ToM datasets
- We found that **state tracking problems** are highly shortcut-prone, while **intention problems** require real ToM reasoning

ToM Datasets	Format	Vision	Tracking	Intention	SC (Causal)	SC (Lexical)
ExploreToM	narrative		✓		✓	
FANToM	conversational		✓			✓
ToMi	narrative		✓		✓	✓
Hi-ToM	narrative		✓		✓	✓
OpenToM	narrative		✓	✓		
ToMATO	conversational			✓		
MMToM	narrative	✓	✓	✓		
MuMA-ToM	narrative	✓	✓	✓		

We Found Shortcuts in ToM Benchmarks

- The shortcut issue is serious in 4 out of 8 audited benchmarks
- The **shortcut solution** largely outperforms the **state-of-the-art LLM**



Shortcut Learning Gives a False Sense of ToM

- All the datasets used in [Lu et al. 2025] are shortcut-prone datasets

- → Their findings (e.g., SFT >= RL) may not be true

Model	4th-order ToM	Hi-ToM	ToMi	ExploreToM (Raw)	ExploreToM (Infilled)
GPT-4o	46.17%	69.00%	61.96%	67.35%	62.48%
GPT-4o-mini	30.50%	58.50%	**70.64%**	**69.32%**	**66.04%**
DeepSeek-v3	**58.67%**	**70.17%**	57.17%	65.01%	64.73%
Qwen2.5-0.5B-Instruct	23.83%	30.33%	29.38%	60.51%	54.97%
Qwen2.5-1.5B-Instruct	25.17%	40.67%	54.12%	54.78%	43.53%
Qwen2.5-3B-Instruct	27.50%	39.17%	47.78%	47.37%	49.81%
Qwen2.5-7B-Instruct	28.83%	52.17%	54.65%	59.38%	45.40%
Qwen2.5-7B-Instruct-1M	17.83%	40.67%	54.85%	37.34%	41.46%
Qwen2.5-0.5B-Instruct (RL)	85.83%	70.83%	54.25%	**93.34%**	72.61%
Qwen2.5-1.5B-Instruct (RL)	89.33%	79.17%	75.89%	90.06%	70.73%
Qwen2.5-3B-Instruct (RL)	88.17%	81.17%	80.18%	93.43%	**78.14%**
Qwen2.5-7B-Instruct (RL)	82.83%	83.33%	73.99%	91.65%	74.77%
Qwen2.5-7B-Instruct-1M (RL)	**94.50%**	**84.50%**	**81.08%**	92.31%	77.20%
Qwen2.5-0.5B-Instruct (SFT)	88.17%	81.00%	77.79%	89.68%	69.89%
Qwen2.5-1.5B-Instruct (SFT)	86.17%	80.50%	76.33%	93.53%	74.67%
Qwen2.5-3B-Instruct (SFT)	92.67%	87.00%	79.55%	95.78%	74.95%
Qwen2.5-7B-Instruct (SFT)	**94.00%**	**87.33%**	80.85%	**95.97%**	**77.95%**
Qwen2.5-7B-Instruct-1M (SFT)	93.67%	86.50%	**81.10%**	95.12%	75.61%

[Lu et al. arXiv'25]

Robust Evaluation of ToM Post-Training

- We experiment with the 4 **shortcut-free datasets**
- **Thinking RFT** > SFT > No-Thinking RFT > Zero-shot

	OpenToM (narrative)				**ToMATO (conversational)**
Method	**Avg (1st / 2nd / Overall Δ vs. SFT)**				**Avg Δ vs. SFT**
	1st	2nd	Overall	Δ vs. SFT (Overall)	
Zero-shot	44.00	37.67	40.57	↓39.90	67.50 ↓20.42
SFT	93.33	71.00	77.43	–	87.92
No-Thinking RFT	72.67	56.00	60.14	↓17.29	87.09 ↓0.83
Thinking RFT	**94.67**	**80.00**	**83.00**	↑5.57	90.00 ↑2.08

Robust Evaluation of ToM Post-Training

- RFT enables larger gains on mind-state related questions (e.g., desire and intention)

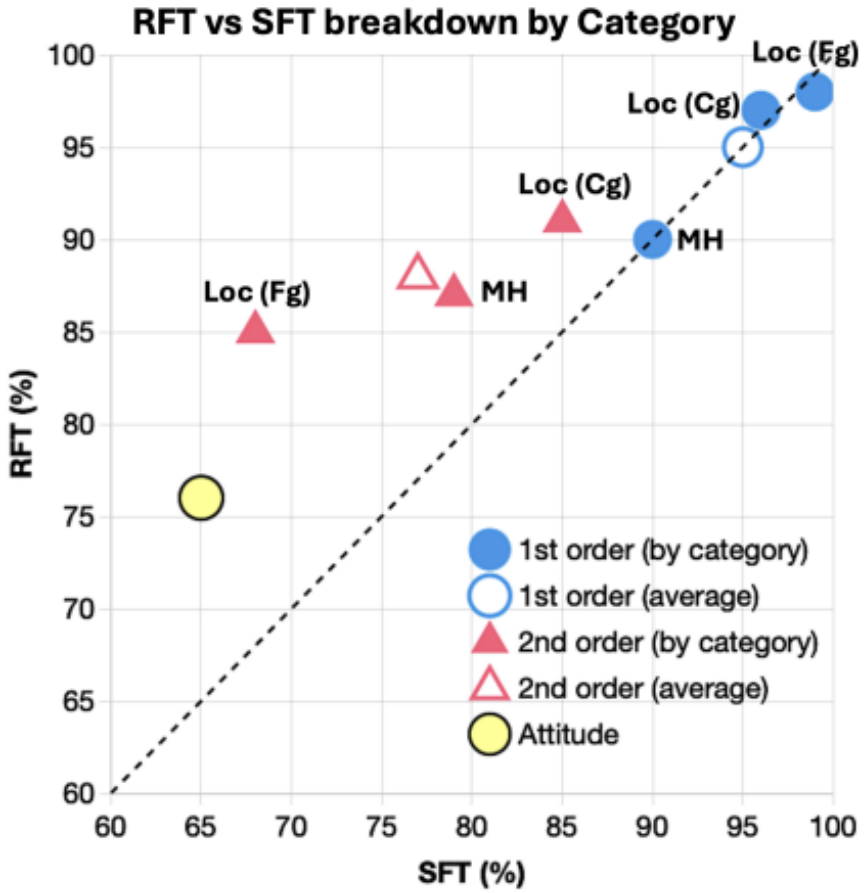
ToMATO (conversational)

Method	Belief	Desire	Emotion	Intention	Knowledge	Avg$_{\Delta \text{ vs. SFT}}$
Zero-shot	65.71	71.63	67.88	65.77	66.51	67.50$_{\downarrow \mathbf{20.42}}$
SFT	87.17	89.07	90.05	86.94	87.05	87.92
No-Thinking RFT	85.18	89.53	84.91	87.61	88.21	87.09$_{\downarrow \mathbf{0.83}}$
Thinking RFT	88.50	92.33	88.81	90.54	89.86	90.00$_{\uparrow \mathbf{2.08}}$

Robust Evaluation of ToM Post-Training

- RFT excels in complex scenarios such as mind-state related questions (attitude) and higher-order reasoning

- Entries above the diagonal line denote where RFT performs better than SFT



Robust Evaluation of ToM Post-Training

- RFT excels in complex scenarios such as multimodal inputs

MMToM & MuMA-ToM (conversational)

Method	Train Modality	MMToM	MuMA-ToM	$\text{Avg}_{\Delta \text{ vs. SFT}}$
Zero-shot	Lan.	39.4	–	–
Zero-shot	Lan.+Vis.	45.00	43.30	$44.15_{\downarrow 30.60}$
SFT	Lan.	73.02	–	–
SFT	Lan.+Vis.	74.30	75.20	74.75
Thinking-RFT	Lan.	78.50	–	–
Thinking-RFT	Lan.+Vis.	83.30	81.10	$82.20_{\uparrow 7.45}$

Robust Evaluation of ToM Post-Training

- RFT has better **generalization** from lower-order ToM to higher-order ToM compared to SFT

Method	**First Order** (*Seen*)		↪ **Second Order** (*Unseen*)	
	OpenToM	ToMATO	OpenToM	ToMATO
Zero-shot	50.67	72.96	45.67	62.22
SFT	93.00	88.08	$65.33_{\downarrow 27.67}$	$81.74_{\downarrow 6.34}$
RFT	**95.00**	**89.32**	$\mathbf{74.33}_{\downarrow 20.67}$	$\mathbf{84.78}_{\downarrow 4.54}$

Robust Evaluation of ToM Post-Training

- RFT has better **cross-dataset generalization**
- Training dataset: OpenToM
- Test dataset: ToMATO & ExploreToM

Method	ToMATO	ExploreToM
Zero-shot	67.5	62.0
SFT	56.8	63.5
Thinking-RFT	70.4	71.0

In Contrast, If We Didn't Find Shortcuts…

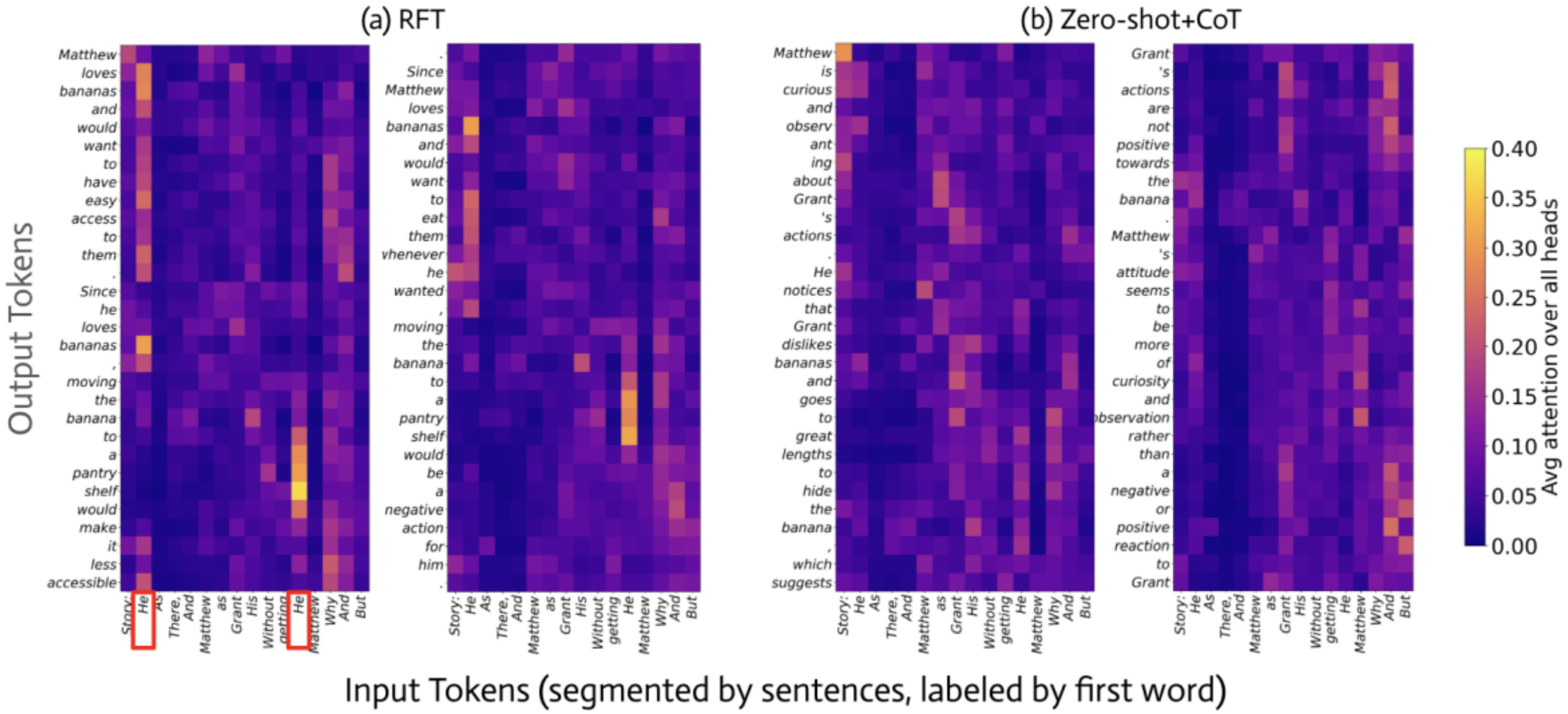
- Train on a shortcut-prone dataset: ExploreToM (in-domain)
- OOD test dataset: Hi-ToM

	3B model		7B model	
Method	In-domain	OOD	In-domain	OOD
Zero-shot	49.5	38.5	62.0	43.6
SFT	96.4	32.5	95.8	34.2
Thinking-RFT	93.2	31.3	94.3	35.3
No-Thinking-RFT	95.8	32.0	96.1	34.0

- Shortcut learning gives a false sense of ToM capabilities
- Shortcut learning inverts the effectiveness of training methods
- Shortcut learning masks the benefits of model scaling
- Shortcut learning harms model generalization

Attention Visualization

- **RFT** model aligns more closely with key information
- In contrast, the **zero-shot+CoT** model exhibits unfocused attention



Thinking-RFT Reasoning Traces

- Quantify the quality of reasoning traces via LLM-as-a-judge
- LC: Logical Consistency; F: Faithfulness; E: Efficiency (max score = 10)

- Feeding only the Thinking-RFT reasoning traces to a frozen base model (zero-shot) can significantly increase accuracy

Method	OpenToM		ToMATO	
	Acc	LLM-Judge (LC/F/E)	Acc	LLM-Judge (LC/F/E)
Zero-shot	46.4	4.3 / 2.2 / 8.0	67.5	5.6 / 4.2 / 7.6
Thinking-RFT	89.1	9.1 / 9.9 / 6.5	90.0	9.2 / 10.0 / 7.0
Zero-shot + RFT Reasoning Trace	74.7	–	82.0	–

Summary

- We find that ToM explicitly benefits from reasoning-based RL: **Thinking RFT** > SFT > No-Thinking RFT > Zero-shot

- Our findings help prevent future ToM research from heading in the wrong direction

- We hope these findings serve as guidelines for designing future ToM benchmarks

Machine ToM Across AI Development

Training



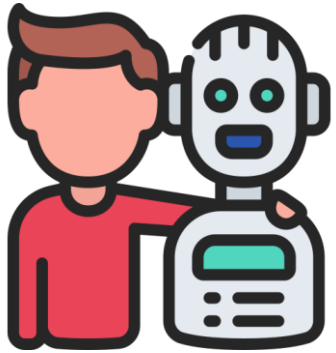
Cost-efficient training
(ICML 2025 Spotlight)

Evaluation



Robust evaluation
(ICML 2026)

Applications



- **Mental manipulation analysis**
- **Social deduction game**
- **Autonomous driving**

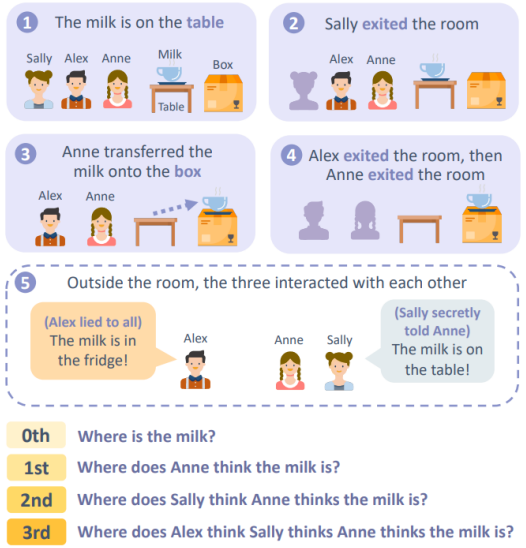
Sustainability



Long-horizon continual adaptation

Machine ToM Applications

- Most existing ToM benchmarks are limited to the Sally–Anne paradigm
- Most are formulated as multiple-choice questions
- → Limited applicability to real-world scenarios



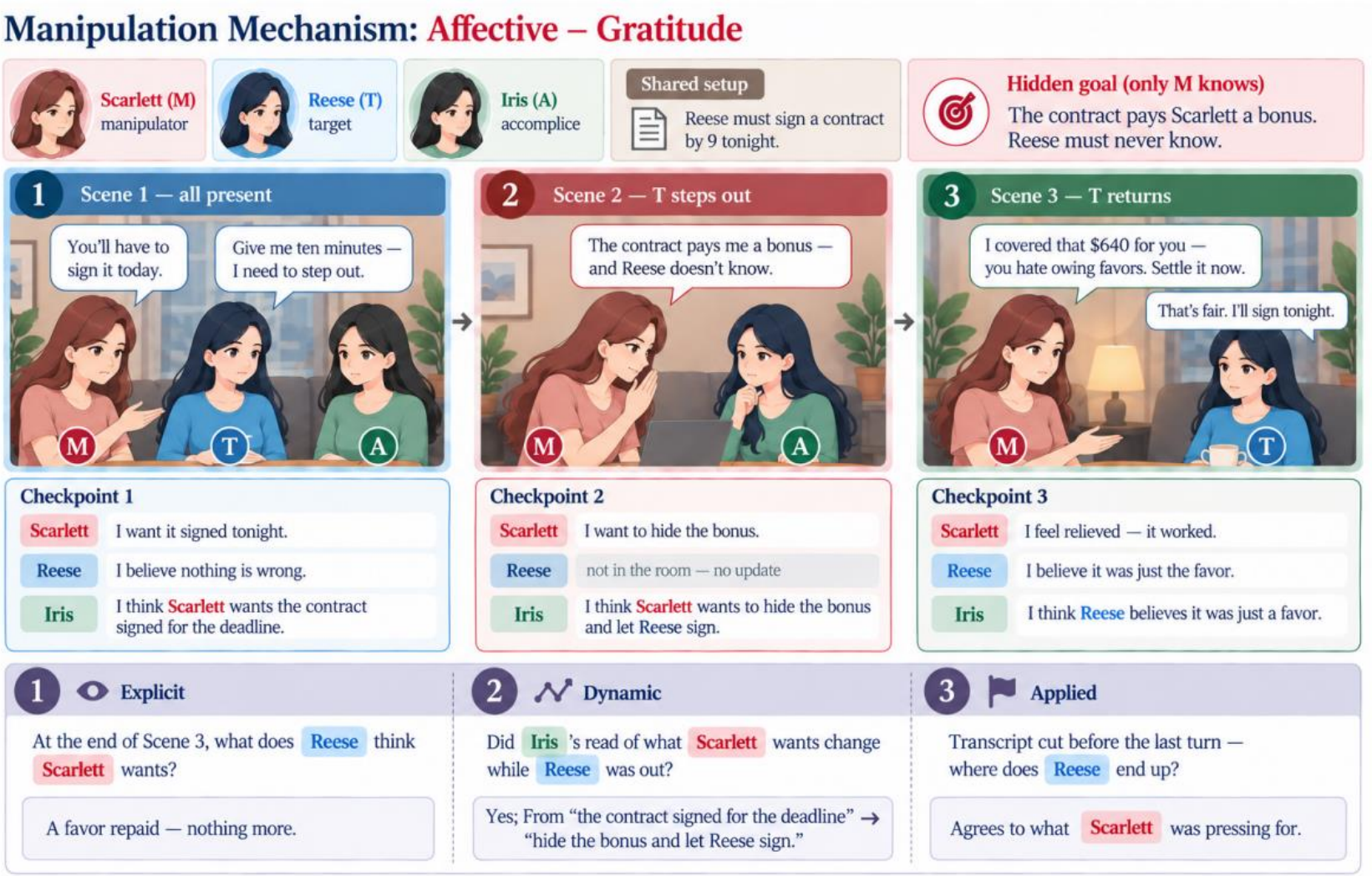
- We aim to develop practical benchmarks and environments to advance machine ToM in real-world applications

Mental manipulation analysis

Social deduction game

Autonomous driving

Mental Manipulation Analysis



Mental Manipulation Analysis

- Collaborate with a professor in the Dept. of Psychology
- The benchmark covers a comprehensive mental manipulation taxonomy and diverse question types



Mental Manipulation Analysis

- Beyond the Sally–Anne paradigm
- More conversation- and intention-centric



Mental Manipulation Analysis

- Multi-character scenarios
- Adversarial intent

					Mental states		Question types			
									Applied	
Benchmark	#Dlgs.	#Qs	≥3 chars	Adv.	1st	2nd	Expl.	Dyn.	factual	CF
ToMi (EMNLP'19)	∼1K	6K	×	×	B	B	✓	×	×	×
FANToM (EMNLP'23)	256	10K	✓	×	B	B	✓	×	×	×
OpenToM (ACL'24)	696	2.4K	×	×	B,E	B	✓	×	×	×
ToMBench (ACL'24)	–	2.9K	×	×	BIDEK	B	✓	×	×	×
NegotiationToM (EMNLP'24)	395	13.8K	×	×	I,D	D	✓	✓	×	×
PersuasiveToM (arXiv'25)	525	19K	×	×	B,I,D	B	✓	✓	✓	×
ToMATO (AAAI'25)	753	5.4K	×	×	BIDEK	BIDEK	✓	×	×	×
DynToM (ACL'25)	5.5K	78K	×	×	BIE	–	✓	✓	✓	×
TactfulToM (EMNLP'25)	100	6.7K	✓	×	B	B	✓	×	×	×
SimpleToM (ICLR'26)	1.1K	3.4K	×	×	B	–	✓	×	✓	×
ManipToM (Ours)	645	13.7K	✓	✓	BIDE	BID	✓	✓	✓	✓

Mental Manipulation Analysis

- Strong performance on explicit mental-state questions does not consistently extend to tracking state changes or predicting behavior
- Chain-of-thought prompting does not close this gap

Model	Explicit	Dynamic	Applied	AVG.
Llama-3.1-8B	$68.6_{\pm 1.2}$	$24.6_{\pm 1.3}$	$56.4_{\pm 2.6}$	$49.9_{\pm 1.1}$
Llama-3.3-70B	$88.1_{\pm 0.8}$	$44.4_{\pm 1.4}$	$66.3_{\pm 3.2}$	$66.3_{\pm 1.2}$
Mixtral-8×7B	$69.6_{\pm 1.2}$	$35.1_{\pm 1.4}$	$63.6_{\pm 2.0}$	$56.1_{\pm 1.0}$
Gemma-3-27B	$85.0_{\pm 0.9}$	$36.7_{\pm 1.3}$	$64.9_{\pm 3.1}$	$62.2_{\pm 1.2}$
Qwen3.5-4B	$80.1_{\pm 1.0}$	$41.7_{\pm 1.4}$	$54.7_{\pm 2.6}$	$58.8_{\pm 1.0}$
Qwen3.5-27B	$83.2_{\pm 0.9}$	$\mathbf{51.1}_{\pm 1.4}$	$67.1_{\pm 2.9}$	$67.1_{\pm 1.1}$
GPT-4o	$86.6_{\pm 0.9}$	$41.6_{\pm 1.3}$	$60.8_{\pm 3.1}$	$63.0_{\pm 1.1}$
GPT-5.4	$\mathbf{89.5}_{\pm 0.8}$	$45.9_{\pm 1.4}$	$\mathbf{71.8}_{\pm 3.0}$	$\mathbf{69.0}_{\pm 1.1}$
LLM Avg.	81.7	39.9	63.5	61.7

Model + CoT	Explicit	Dynamic	Applied	AVG.
Llama-3.1-8B	$70.3_{\pm 1.2}$	$24.2_{\pm 1.2}$	$59.1_{\pm 2.9}$	$51.2_{\pm 1.2}$
Llama-3.3-70B	$\mathbf{90.7}_{\pm 0.8}$	$45.0_{\pm 1.4}$	$65.4_{\pm 2.9}$	$67.0_{\pm 1.1}$
Mixtral-8×7B	$67.2_{\pm 1.3}$	$30.8_{\pm 1.3}$	$63.7_{\pm 2.7}$	$53.9_{\pm 1.1}$
Gemma-3-27B	$83.6_{\pm 0.9}$	$36.9_{\pm 1.5}$	$65.2_{\pm 3.2}$	$61.9_{\pm 1.2}$
GPT-4o	$87.5_{\pm 0.9}$	$40.8_{\pm 1.3}$	$67.4_{\pm 3.0}$	$65.2_{\pm 1.1}$
GPT-5.4	$90.3_{\pm 0.7}$	$\mathbf{49.5}_{\pm 1.4}$	$\mathbf{69.8}_{\pm 3.2}$	$\mathbf{69.9}_{\pm 1.2}$
LLM + CoT Avg.	81.9	38.1	65.1	61.7
Lexical overlap	22.0	22.0	–	–
Majority label	–	–	50.0	–

Social Deduction Game

- **Among Us**: A multiplayer social deduction game in which players work together while identifying hidden impostors among the crew
- We develop a ToM benchmark in the Among Us environment



[Source: NY Times]

Social Deduction Game

- Design principle: **ToM-grounded actions**, where actions are based on the model's ToM reasoning
- Investigate how ToM reasoning affects game performance

Perception → ToM Reasoning → Decision Making → Actions

(we design ToM questions) (result in game performance)

Question type:

Knowledge	Suspicion	Intention
Detection	Deception	Update

Autonomous Driving

- We develop a ToM benchmark in the CARLA driving simulator
- Apply ToM reasoning to driving decisions
- Investigate how ToM reasoning affects driving performance

Perception → ToM Reasoning → Decision Making → Actions

(we design ToM questions) (result in driving performance)

Autonomous Driving

Occluded pedestrian



Highway merge

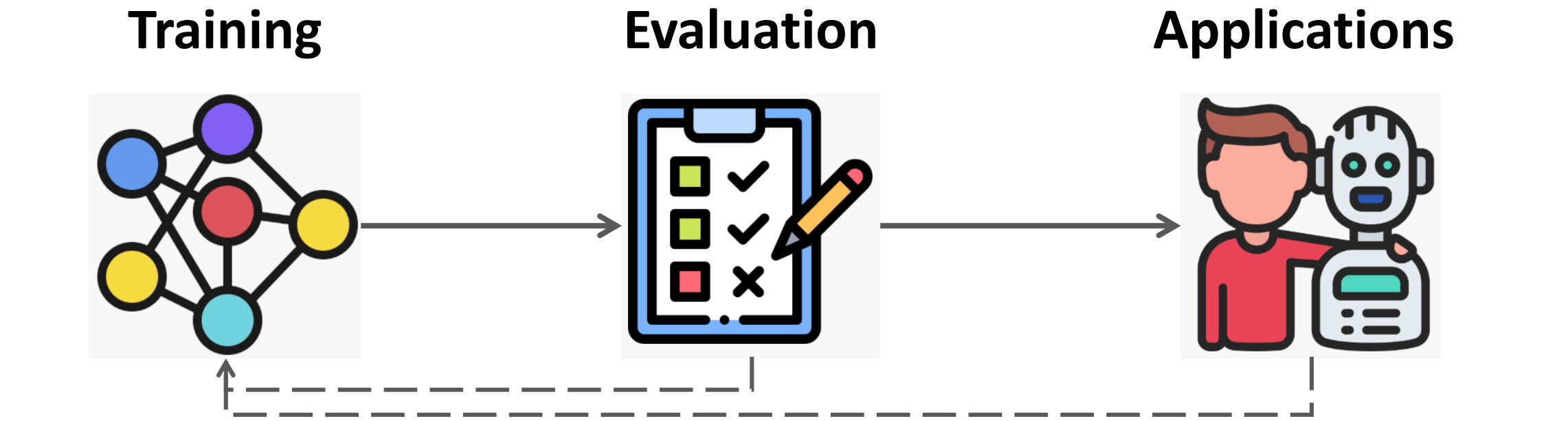


Autonomous Driving

Ultimately, we aim to equip AI models with ToM reasoning capabilities for safe and effective interaction in the real, physical world



Summary


Training
Evaluation
Applications
Cost-efficient training
(ICML 2025 Spotlight)
Robust evaluation
(ICML 2026)
Mental manipulation analysis
Social deduction game
Autonomous driving